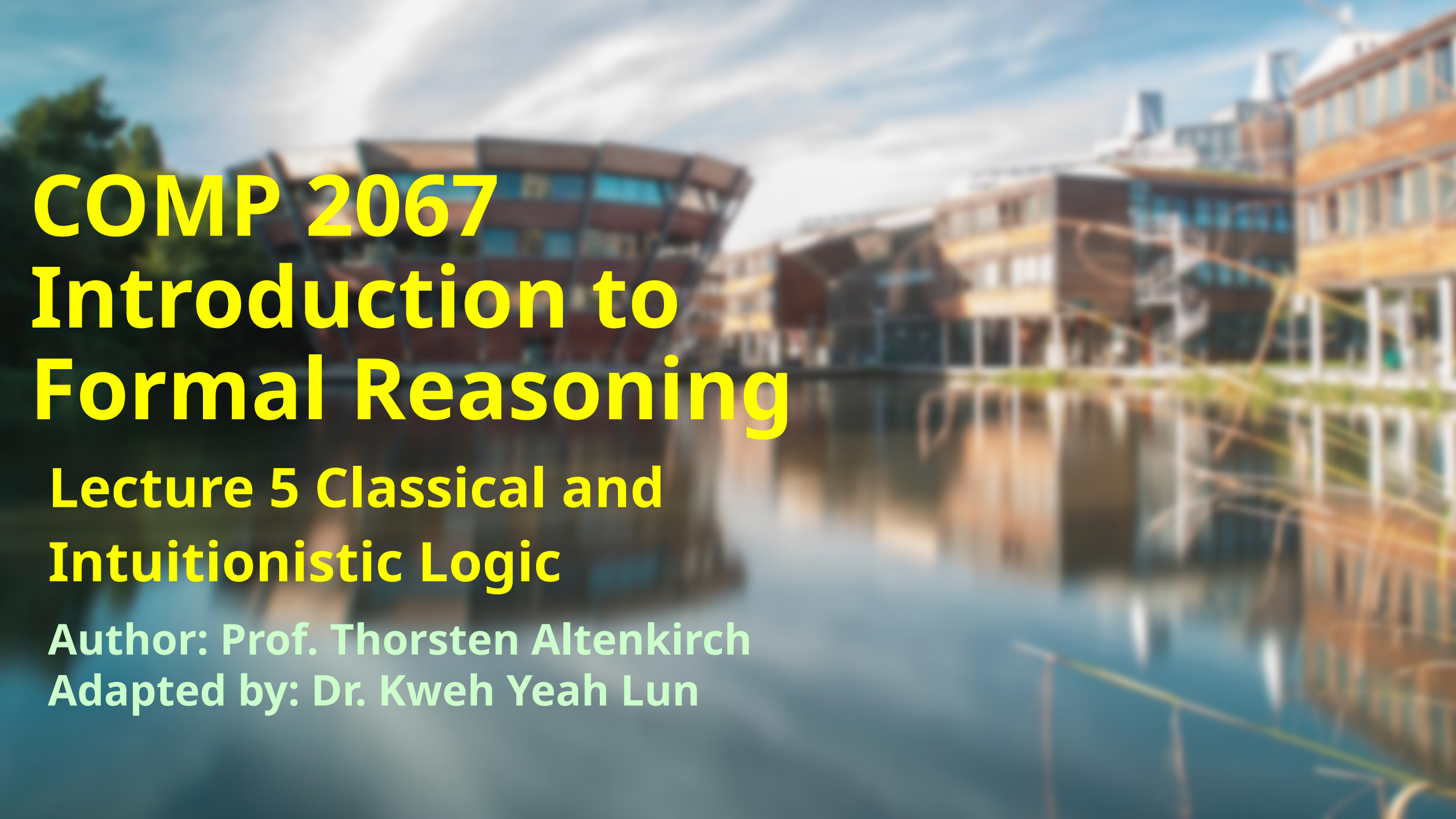




COMP 2067

Introduction to

Formal Reasoning

Lecture 5 Classical and

Intuitionistic Logic

Author: Prof. Thorsten Altenkirch
Adapted by: Dr. Kweh Yeah Lun



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Nottingham

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Learning Outcomes

At the end of this lecture, you will be able

- to use Lean to prove De-Morgan's Law
- to elaborate the concept of EM Law and RAA
- to understand proof by contradiction
- to apply EM Law and RAA in Lean
- to elaborate on the difference between classical logic and intuitionistic logic



The de Morgan Laws

- $\neg (P \vee Q) \leftrightarrow \neg P \wedge \neg Q$
- $\neg (P \wedge Q) \leftrightarrow \neg P \vee \neg Q$
- **Exercise 1:** Let's try to prove these laws using the truth table.



- Example: De Morgan's 1st Law

Try it!

```
theorem dm1 : ¬ (P ∨ Q) ↔ ¬ P ∧ ¬ Q :=  
begin  
  constructor,  
  assume npq,  
  constructor,  
  assume p,  
  apply npq,  
  left,  
  exact p,  
  assume q,  
  apply npq,  
  right,  
  exact q,  
  assume npnq pq,  
  cases npnq with np nq,  
  cases pq with p q,  
  apply np,  
  exact p,  
  apply nq,  
  exact q,  
end
```



- Example: De Morgan's 2nd L

[Try it!](#)

```
theorem dm2 : ¬ (P ∧ Q) ↔ ¬ P ∨ ¬ Q :=  
begin  
  constructor,  
  assume npq,  
  left,  
  assume p,  
  apply npq,  
  constructor,  
  exact p,  
  sorry,  
  assume npnq pq,  
  cases pq with p q,  
  cases npnq with np nq,  
  apply np,  
  exact p,  
  apply nq,  
  exact q,  
end
```



- Right to left is fine, but **left to right, get stuck.**

```
npq : ¬(P ∧ Q)  
⊢ ¬P ∨ ¬Q
```

- If go **left**, will end up with:

```
npq : ¬(P ∧ Q)  
⊢ ¬P
```

- Then, **assume p**:

```
npq : ¬(P ∧ Q),  
p : P  
⊢ false
```

- Then **apply npq**:

```
npq : ¬(P ∧ Q),  
p : P  
⊢ P ∧ Q
```



- Then **constructor, exact p**.
- Now, end up with:

```
npq : ¬(P ∧ Q),  
p : P  
⊢ Q
```

- It is an unprovable goal.
- Let's consider this **example**:

It is not the case that I have a laptop and that I have a desktop.

Can you conclude that

I don't have a laptop or I don't have a desktop?

No, because you don't know which one is the case, that is you don't have

evidence for either.



- "It is not the case that I have a laptop and that I have a desktop."
- This is a **negation of a conjunction**.
- Let's translate it into simpler terms:
- **Conjunction:** "I have a laptop **AND** I have a desktop." (**Both must be true**)
- **Negation:** "It is not true that both are true."
- In other words, the original statement tells us that **you don't have both a laptop and a desktop simultaneously.**



- The **Proposed Conclusion**:
- "I don't have a laptop **OR** I don't have a desktop."
- This is a **disjunction** (using "or").
- It means **at least one of these must be true**:
 - You don't have a laptop.
 - You don't have a desktop.
 - You don't have either.



- **Why the Conclusion Doesn't Follow**
- The key is that the original statement only tells us that you **don't have both**.
- It **doesn't specify which one (or if either) you lack**.
- **Scenario 1: You have a laptop but not a desktop.**
 - This satisfies the original statement ("not both") but doesn't fit the conclusion ("don't have a laptop **or** don't have a desktop").
- **Scenario 2: You have a desktop but not a laptop.**
 - Same as above, the original is true, but the conclusion isn't fully accurate.
- **Scenario 3: You have neither a laptop nor a desktop.**
 - This satisfies both the original statement and the conclusion.



- The **original statement** is **less specific** than the proposed conclusion.
- It only rules out the case where you have both a laptop and a desktop.
- It doesn't provide enough information to definitively say you lack either one specifically.



The law of the excluded middle (EM)

- **Axiom: The law of the excluded middle**
Every proposition is either true or false.
- That is: For any proposition P : $P \vee \neg P$
- In latin this law is called ***Tertium non datur***, which translates to *the 3rd is not given*.
- In Lean, this axiom can be accessed by

```
open classical  
  
#check em P
```

Try it!

- Command: **#check** -> checks the type of a term



- For any proposition P , $\text{em } P$ proves $P \vee \neg P$.
- **Example: De Morgan's 2nd Law – Left to right** direction using $\text{em } P$

```
theorem dm2_em : ¬ (P ∧ Q) → ¬ P ∨ ¬ Q :=  
begin  
  assume npq,  
  cases em P with p np,  
  right,  
  assume q,  
  apply npq,  
  constructor,  
  exact p,  
  exact q,  
  left,  
  exact np,  
end
```

Try it!



Principle of Indirect Proof (raa)

- This principle states that to prove P it is sufficient to show that $\neg P$ is impossible.
- In latin, it is known as *reduction ad absurdo* (reduction to the absurd).



- **Relationship between Double negation rule and Reductio ad Absurdum (RAA)**
- Assume $\neg P$: In RAA, start by assuming the opposite: prove $(\neg P)$ is true.
- Derive a Contradiction: try to derive a contradiction from this assumption.
- Double Negation: If reach a contradiction, you've effectively shown that $\neg P$ leads to something impossible.
- This can be expressed as $\neg\neg P$ (not not P).
- Apply Double Negation Elimination: use the rule $\neg\neg P \rightarrow P$ to conclude that P must be true.
- If not not P, then P
- If P is not false, then P is true.



Example of Proof by Contradiction: Prove Your Partner Truly Loves You

- 1.The Claim:** P: "My partner truly loves me."
- 2.Assume the Opposite ($\neg P$):** "My partner does not truly love me."
- 3.Derive a Contradiction:** You can't find the evidence that your partner does not truly love you. So, there is a **contradiction**.
- 4.Conclusion:** Since the assumption that your partner doesn't love you leads to a contradiction, the **assumption must be false**.
- 5.Therefore,** your original claim – "My partner truly loves me" is **true**.



- **Example:** Derive **raa** using **em**
Strategy: Proof by contradiction

```
theorem raa :  $\neg \neg P \rightarrow P$  :=  
begin  
  assume nnp,  
  cases (em P) with p np,  
  exact p,  
  have f : false,  
  apply nnp,  
  exact np,  
  cases f,  
end
```

Try it!



- The idea is to assume $\neg\neg P$ and then prove P by analysing $P \vee \neg P$.
- In the case P , done.
- In the case $\neg P$, we have a contradiction with $\neg\neg P$ and we can use that false implies everything.
- `have f : false`
 - The tactic `have` introduces a variable `f` with the type `false`
 - current goal is split into two subgoals:
 - one is to prove the hypothesis asserted, which is `false`,
 - and the other is to continue the original proof assuming that hypothesis, which is `f : false`.



- Example: Derive em from raa

Prove $\neg \neg (P \vee \neg P)$ without using classical logic

```
theorem nn_em :  $\neg \neg (P \vee \neg P)$  :=  
begin  
  assume npnp,  
  apply npnp,  
  right,  
  assume p,  
  apply npnp,  
  left,  
  exact p,  
end
```

[Try it!](#)



- After **apply npnp**

```
P : Prop,  
npnp : ¬(P ∨ ¬P)  
⊢ P ∨ ¬P
```

- Let's go **right**.
- After a few steps we are in the same situation
- But something has changed.
- This time go **left** and it is done.

```
P : Prop,  
npnp : ¬(P ∨ ¬P),  
p : P  
⊢ P ∨ ¬P
```



- If assume we have a **constant raa**, we can show **em**.
- Examples:

```
constant raa :  $\neg\neg P \rightarrow P$   
  
theorem em :  $P \vee \neg P :=$   
begin  
  apply raa,  
  apply nn_em,  
end
```

[Try it!](#)



- The **constant** tactic is used to **introduce new constants or axioms** into the environment.
- It declares a new symbol along with its type without providing any definition or proof for it.
- It's important to **use constant with caution**, as introducing arbitrary axioms can potentially lead to inconsistencies or invalid proofs if those axioms are contradictory or incompatible with other established facts.



Intuitionistic and Classical Logic

- **Intuitionistic logic emphasizes constructive proofs**, proving existence by providing an actual example or method satisfying the proposition.
- It **rejects the Law of Excluded Middle (LEM)**, meaning it **doesn't assume every statement is either true or false**.
- **Double negation elimination ($\neg\neg P \vdash P$) isn't always valid**.
- Need a constructive proof even after negating twice.



- Examples of constructive proofs:
 - Proving that there exist even and odd natural numbers:
 - Instead of using a proof by contradiction, a constructive proof would show how to build even and odd numbers starting from 0 (e.g. 2 is constructed by adding 1 to 1, and 3 is constructed by adding 1 to 2).
 - Proving that the square root of 2 exists:
 - A constructive proof might present an algorithm for approximating the square root of 2 with increasing accuracy using an iterative process.



- Applications of constructive proofs:
 - Computer science: Verification of programs, design of algorithms, and type theory
 - Mathematics: Constructive mathematics emphasizes proofs that build objects explicitly, leading to deeper insights into their properties and potential uses.
 - Foundations of mathematics: Some mathematicians prefer intuitionistic logic and constructive proofs due to their emphasis on practical realizability.



Classical Logic

- Uses **non-constructive proofs**:
 - often establish existence indirectly, sometimes by showing that the non-existence of the object would lead to a contradiction.
 - can prove a statement exists without providing a specific example or method for finding it.
 - For example, proving that the equation $x^2 + 1 = 0$ has no rational solutions is non-constructive (using proof by contradiction), but you wouldn't necessarily know how to find its real solutions from such a proof.



- Classical logic is also known as **standard logic or Frege-Russell logic**
- **Bivalence:** Every proposition is either true or false.
 - There are no "third truth values" like "unknown" or "indeterminate."
 - known as the **Law of Excluded Middle (EM)**.
- **Non-contradiction:** A proposition cannot be both true and false at the same time.
- **Double negation elimination:** If the negation of the negation of a proposition is true, then the proposition itself is true.
 - This means "not not P" logically implies "P".
- provides various rules and principles for drawing valid conclusions from given propositions, such as modus ponens, modus tollens, and syllogism.



- Should we always assume **em** (or alternatively **raa**), hence should we always work in classical logic?
- Intuitionistic logic is constructive, and it is important to computer scientists: whenever we show that something exists, we are actually able to compute it.



- Give yourself a big clap.
- Survived once again? Great! ◀◀
- Any question?



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Next Lecture...

- Predicate logic



“The Web as I envisaged it, we have not seen it yet. The future is still so much bigger than the past.”

– Tim Berners-Lee.

Hope you all enjoy this lecture!